

Van der Waalsova enačba

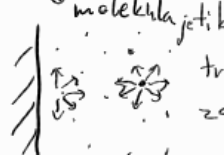
$$V_m = \frac{VM}{m} = \frac{M}{\rho}$$

idealni plin: $pV_m = RT$

neidealni plin: $(p + \frac{a}{V_m^2})(V_m - b) = RT$

molekule imajo lastno V , zato so ostale na peti in ne morejo zavzeti celotnega prostora

molekule jstik pred trkom zavirata



1.1 Prepisi VdW enačbo v brezdimenzijsko obliko

$$P = \frac{p}{p_c}, \quad V = \frac{V_m}{V_{m,c}}, \quad T = \frac{T}{T_c}$$

- v krit. točki $(\frac{\partial p}{\partial V_m})_T = 0; (\frac{\partial^2 p}{\partial V_m^2})_T = 0$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$(\frac{\partial p}{\partial V_m})_T = -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3} \stackrel{\text{k.T.}}{=} 0 \Rightarrow$$

$$\frac{V_{m,c} - b}{2} = \frac{V_{m,c}}{3} \Rightarrow V_{m,c} = 3b$$

$$(\frac{\partial^2 p}{\partial V_m^2})_T = \frac{2RT}{(V_m - b)^3} - \frac{6a}{V_m^4} \stackrel{\text{k.T.}}{=} 0$$

$$\frac{RT_c}{4b^2} = \frac{2a}{27b^3}$$

$$p_c = \frac{R \cancel{8a} 4}{27 \cancel{4} b \cdot \cancel{4} b} - \frac{a}{9b^2} =$$

$$T_c = \frac{8a}{27Rb}$$

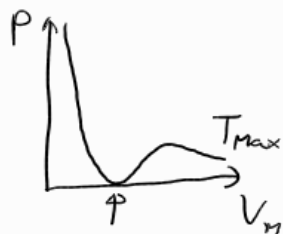
$$p_c = \frac{a}{27b^2}$$

$$\Rightarrow (P p_c + \frac{a}{(V V_{m,c})^2})(V V_{m,c} - b) = RT T_c$$

$$(P \frac{a \cdot 1}{27b^2} + \frac{a \cdot 1 \cdot 3}{V^2 9b^3})(3Vb - b) = R T \frac{8a}{27Rb} \Rightarrow (P + \frac{3}{V^2})(3V - 1) = 8T$$

1.3) $T_{\max}$ pregrete vDW tekočine pri majhnem tlaku?
 $p \ll p_c \rightarrow p \approx 0$

→ iščemo izotermo:



$$\left(\frac{\partial P}{\partial V_m} \right)_T = 0$$

$$p(V_{\max}, T_{\max}) = 0$$

↓ greno v brezdim.

$$\left(\frac{\partial P}{\partial V} \right)_T = 0$$

$$P = 0$$

$$P = \frac{8T}{3V-1} - \frac{3}{V^2} = 0$$

$$\frac{\partial P}{\partial V} = -\frac{8T3}{(3V-1)^2} + \frac{6}{V^3} = 0$$

√ ÷

$$\frac{3V-1}{3} = \frac{V}{2}$$

$$V_{\max} = \frac{2}{3}$$

$$\frac{8T}{1} = \frac{3 \cdot 9}{4}$$

$$T_{\max} = \frac{27}{32} \Rightarrow T_{\max} = \frac{27}{32} T_c$$

1.9) Prožna vrstica (F, l, T sistem)

$$F = A \cdot \alpha \cdot T \left(\frac{l}{l_0} - [1 + \delta(T - T_0)] \left(\frac{l}{l_0} \right)^2 \right)$$

$A, \alpha, l_0, T_0, \delta$ konst.
 $\varnothing$ presek

a) $\alpha = ?$

b) izotermni prožnostni modul

$$\alpha = \frac{1}{l} \left(\frac{\partial l}{\partial T} \right)_F$$

$$E_T = \frac{l}{A} \left(\frac{\partial F}{\partial l} \right)_T$$

$$E_T = \frac{l}{A} A \alpha T \left(\frac{1}{l_0} + 2[1 + \delta(T - T_0)] \frac{l_0^2}{l^3} \right)$$

a) $dF = \left(\frac{\partial F}{\partial T} \right)_l dT + \left(\frac{\partial F}{\partial l} \right)_T dl = 0$ weil $F = \text{konst.}$

$$\frac{dl}{dT} = - \frac{\left(\frac{\partial F}{\partial T} \right)_l}{\left(\frac{\partial F}{\partial l} \right)_T} = - \frac{A \alpha \left(\frac{1}{l_0} - [1 + \delta(2T - T_0)] \frac{l_0^2}{l^3} \right) + A \alpha T \left(- \delta \left(\frac{l_0^2}{l^3} \right) \right)}{A \alpha T \left(\frac{1}{l_0} + 2[1 + \delta(T - T_0)] \frac{l_0^2}{l^3} \right)}$$

$$\alpha = \frac{\frac{1}{l_0} - [1 + \delta(2T - T_0)] \frac{l_0^2}{l^3}}{T \left(\frac{1}{l_0} + 2[1 + \delta(T - T_0)] \frac{l_0^2}{l^3} \right)}$$